

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 6 — Work and Energy

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions × 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions × 5 marks each. Answer *either* the main question *or* the OR alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	Work is done when a force displaces a body. The SI unit of work is:	Newton	Watt	Joule	Pascal	A B C D
2	A man pushes a wall with great force but the wall does not move. The work done on the wall is:	Positive	Negative	Zero	Infinite	A B C D
3	A ball weighing 5 N is lifted to a height of 5 m. The potential energy stored in it is:	100 J	25 J	50 J	250 J	A B C D
4	The kinetic energy of an object depends on its mass and the square of its:	Height	Speed	Weight	Pressure	A B C D
5	A 4 kg body thrown upward at 5 m/s: its KE just before hitting the ground (friction neglected) is:	25 J	50 J	75 J	100 J	A B C D
6	The law of conservation of energy states that energy can neither be created nor destroyed; it can only:	Be lost to heat	Change forms	Increase with time	Decrease with friction	A B C D
7	Power is the rate of doing work. Its SI unit is the:	Joule	Newton	Watt	Horsepower	A B C D
8	100 J of work is done in 5 s. The power utilized is:	10 W	20 W	105 W	500 W	A B C D
9	One horsepower is equal to:	756 W	716 W	736 W	746 W	A B C D
10	Which of the following is a <i>renewable</i> energy resource?	Coal	Natural gas	Sunlight	Uranium	A B C D
11	A petrol engine takes 400 J as input and gives 120 J of useful work. Its efficiency is:	10%	20%	30%	40%	A B C D
12	A practical engine can never have an efficiency equal to or greater than:	0	0.5	0.8	1 (100%)	A B C D

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Define work with its SI unit. Write $W = Fd$. State two conditions when work done is zero.	3	OR	Muhammad Taha pushes a box with 150 N through 5 m on a frictionless surface. Calculate the work done . (Show full working.)	3
	Define kinetic energy . Write the formula $E_k =$			Babar Azam gives a ball (mass 120 g) a kinetic	

(ii)	$\frac{1}{2}mv^2$ and name each variable. Is kinetic energy scalar or vector?	3	OR	energy of 50 J. Calculate its speed .	3
(iii)	Define gravitational potential energy . Write $E_p = mgh$ and explain each symbol. Give two real-life examples.	3	OR	At a sports gala, Kamil (mass 60 kg) jumps 1.5 m high. Calculate his gravitational PE at the top. ($g = 9.8 \text{ m/s}^2$)	3
(iv)	State the Law of Conservation of Energy . Using a falling ball, explain how KE and PE interchange.	3	OR	A football (0.45 kg) is kicked upward at 22 m/s. Find the maximum height it reaches. ($g = 9.8 \text{ m/s}^2$; no friction)	3
(v)	Explain energy conversion with two examples: (a) electricity generation by a dam and (b) a cyclist going uphill.	3	OR	Why is energy always dissipated during conversion? In a petrol car only 17-21% of chemical energy moves the car. Where does the rest go? What is it called?	3
(vi)	Differentiate between renewable and non-renewable energy resources. Give two examples of each.	3	OR	What is a perpetual motion machine ? Why is it impossible? Name the two laws of thermodynamics it would violate.	3
(vii)	Describe how fossil fuels are formed. Name the three main types and write the combustion equation for methane (CH_4).	3	OR	Explain nuclear fuel . Name the two materials used. State one advantage and one disadvantage of nuclear energy.	3
(viii)	Explain hydroelectric generation . What energy conversion takes place? Why can large reservoirs harm the ecosystem?	3	OR	Describe solar radiation as an energy source. State two methods of harnessing it. Give one advantage and one disadvantage.	3
(ix)	Define power . Write $P = W/t$ and state its SI unit. Convert 2 hp to watts. (1 hp = 746 W)	3	OR	A Tesla car engine does $3.69 \times 10^7 \text{ J}$ of work in 1 minute. Find its power in watts and convert to hp. (1 hp = 746 W)	3
(x)	Define efficiency . Write $\eta = E_{\text{out}}/E_{\text{in}} \times 100\%$. Why can no practical machine have 100% efficiency?	3	OR	A 2 hp motor lifts a 10 kg mass to 10 m. Input energy = 2238 J. Find output energy and efficiency . ($g = 9.8 \text{ m/s}^2$)	3
(xi)	A car has kinetic energy E_k . By what factor does KE change if velocity is doubled? Show mathematical reasoning.	3	OR	A 5 kg steel ball is dropped from a 12 m tower. Find its KE on hitting the ground and the speed at impact. ($g = 9.8 \text{ m/s}^2$; no air resistance)	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Define work and its SI unit. State two zero-work conditions. Derive $E_k = \frac{1}{2}mv^2$ using a graphical argument. State the work-energy theorem . A 2.0 kg rock drops from 20 m; after falling 15 m, find its KE and remaining PE. ($g = 9.8 \text{ m/s}^2$)	5	OR	Define gravitational PE and derive $E_p = mgh$. A 5 kg ball drops from 12 m. (a) Find PE at top. (b) Find KE and speed at ground (no friction). (c) If only 540 J of KE at ground, find work done against air resistance. ($g = 9.8 \text{ m/s}^2$)	5
3(ii)	State the Law of Conservation of Energy ($E_{\text{Ti}} = E_{\text{Tf}}$). Show $E_k + E_p = \text{constant}$ for a falling ball. A football (0.45 kg) kicked at 22 m/s: (a) find max height; (b) if friction limits rise to 20 m, find work done against air resistance. ($g = 9.8 \text{ m/s}^2$)	5	OR	Explain energy conversion and dissipation . Give three textbook examples (dam, cyclist, car engine). In a petrol car, only 17-21% moves the car; where does the 80% go? Why are electric cars more efficient? Why can energy never be 100% converted to useful work?	5
3(iii)	Define power ; write $P = W/t$. Explain $1 \text{ W} = 1 \text{ J s}^{-1}$. Define kWh and prove $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. A 1960 W weight lifter raises mass M to 2 m in 3 s. Find M and convert 1960 W to hp. ($g = 9.8 \text{ m/s}^2$; 1 hp = 746 W)	5	OR	Define efficiency ; write $\eta = E_{\text{out}}/E_{\text{in}} \times 100\%$. Why is an incandescent bulb only 5% efficient while a fluorescent lamp is 20%? Why is 100% efficiency impossible? A 2 hp motor lifts 10 kg to 10 m in 1.5 s. Find (a) E_{in} , (b) E_{out} , (c) efficiency. (1 hp = 746 W, $g = 9.8 \text{ m/s}^2$)	5
3(iv)	Differentiate renewable and non-renewable resources (table, 3 examples each). Describe fossil fuels : formation, three types, extraction, combustion equation for CH_4 . State two environmental problems. Why must society shift to renewables?	5	OR	Describe three renewable energy resources : (a) Hydroelectric, (b) Solar, (c) Wind. For each: energy conversion, how electricity is generated, one advantage, one disadvantage. A 8.13 kW pump raises 2500 kg of water 100 m. Find time taken. ($g = 9.8 \text{ m/s}^2$)	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 6 — Work and Energy

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